Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

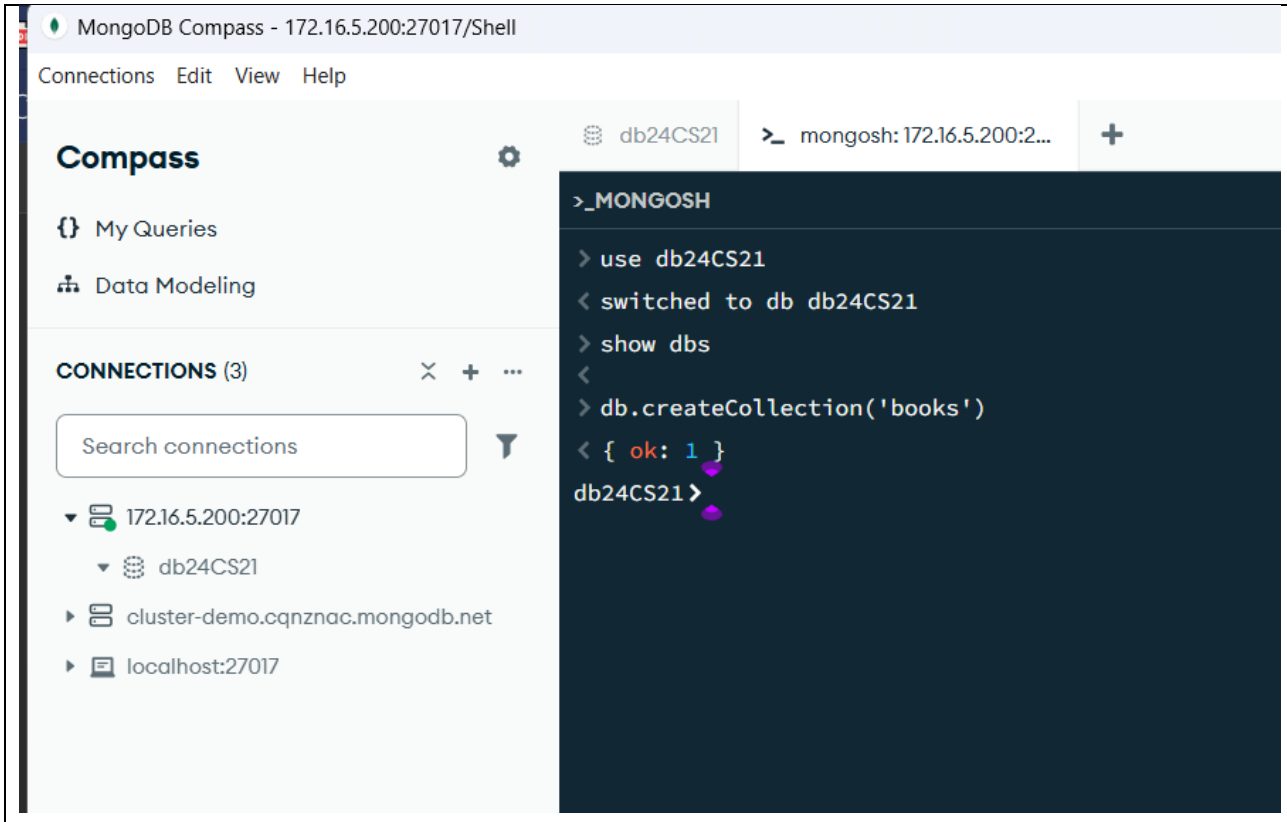
Date: 18 / 02 / 2026

Laboratory Manual

Academic Year	:	2025-26	Semester	:	4
Course code	:	ITUC202	Course name	:	FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

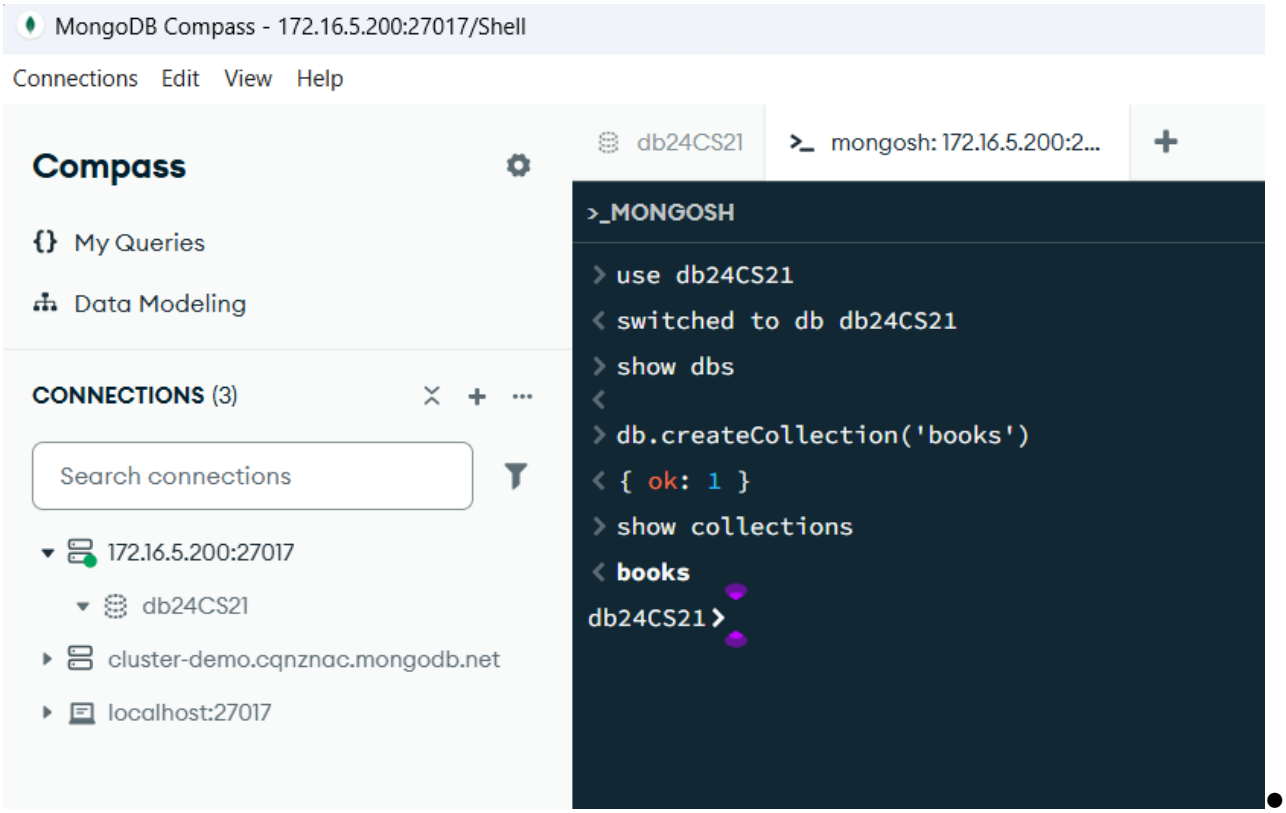
Practical - 5
Aim:- To learn and understand the use of mongodb to create and analyze the data.

To connect to a MongoDB deployment running on localhost with default port 27017, run mongosh without any options:
mongo
OR
mongosh
55
● To show list of data bases available in the system use following command
> show dbs
admin 0.000GB
config 0.000GB
local 0.000GB
● Create a New Database
use <dbname>
> use product
switched to db product
● To display the name of current working Database
> db
product
● To Create a Collection



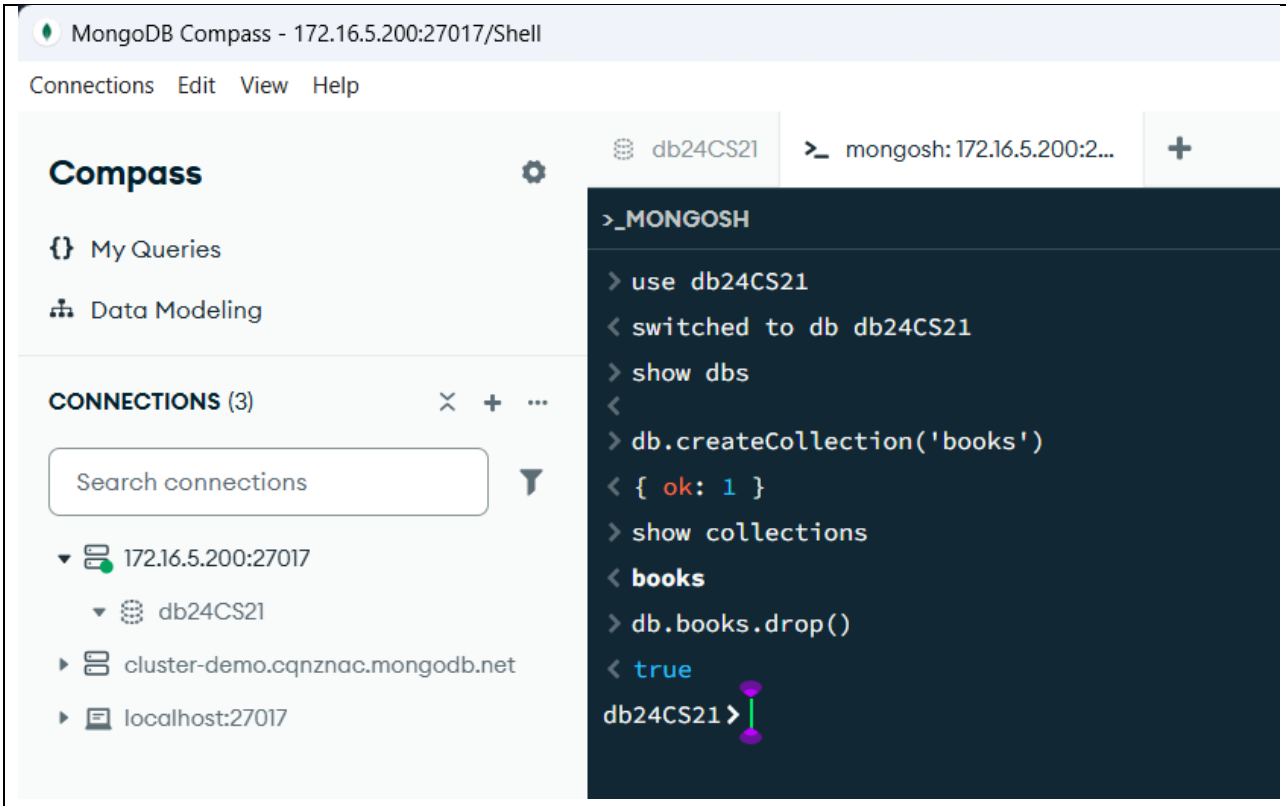
- To Display Collections

> show collections



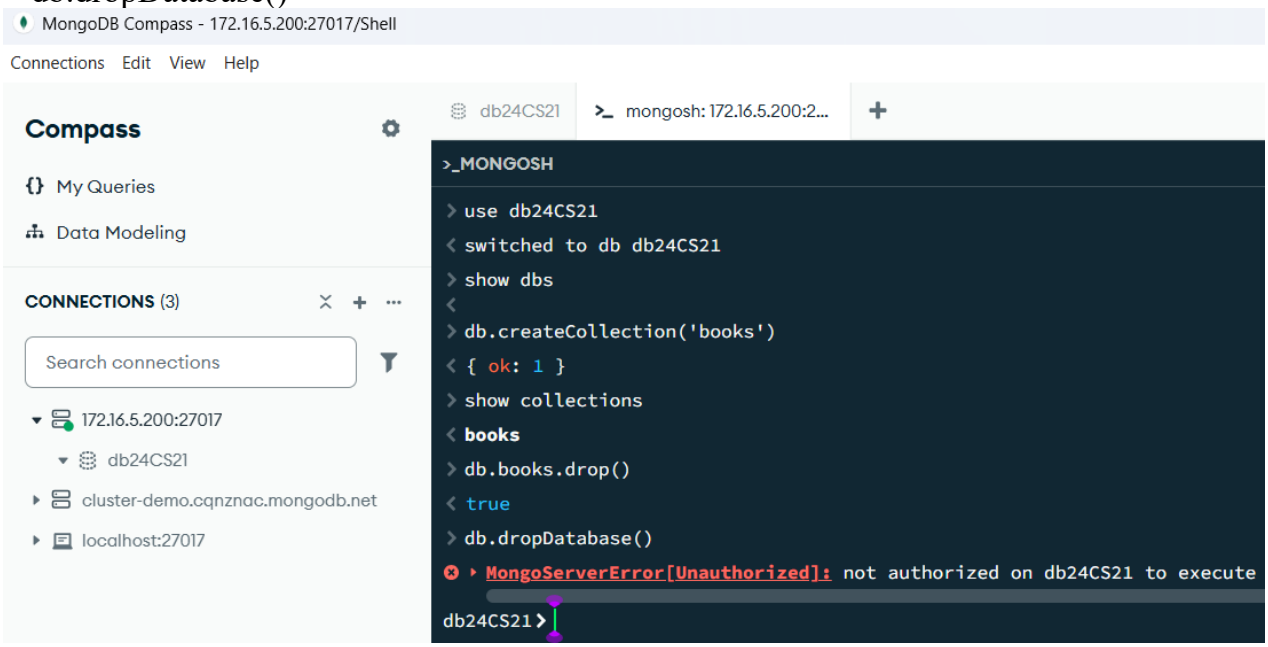
- To Drop a Collection

> db.users.drop()



● To Drop a Database

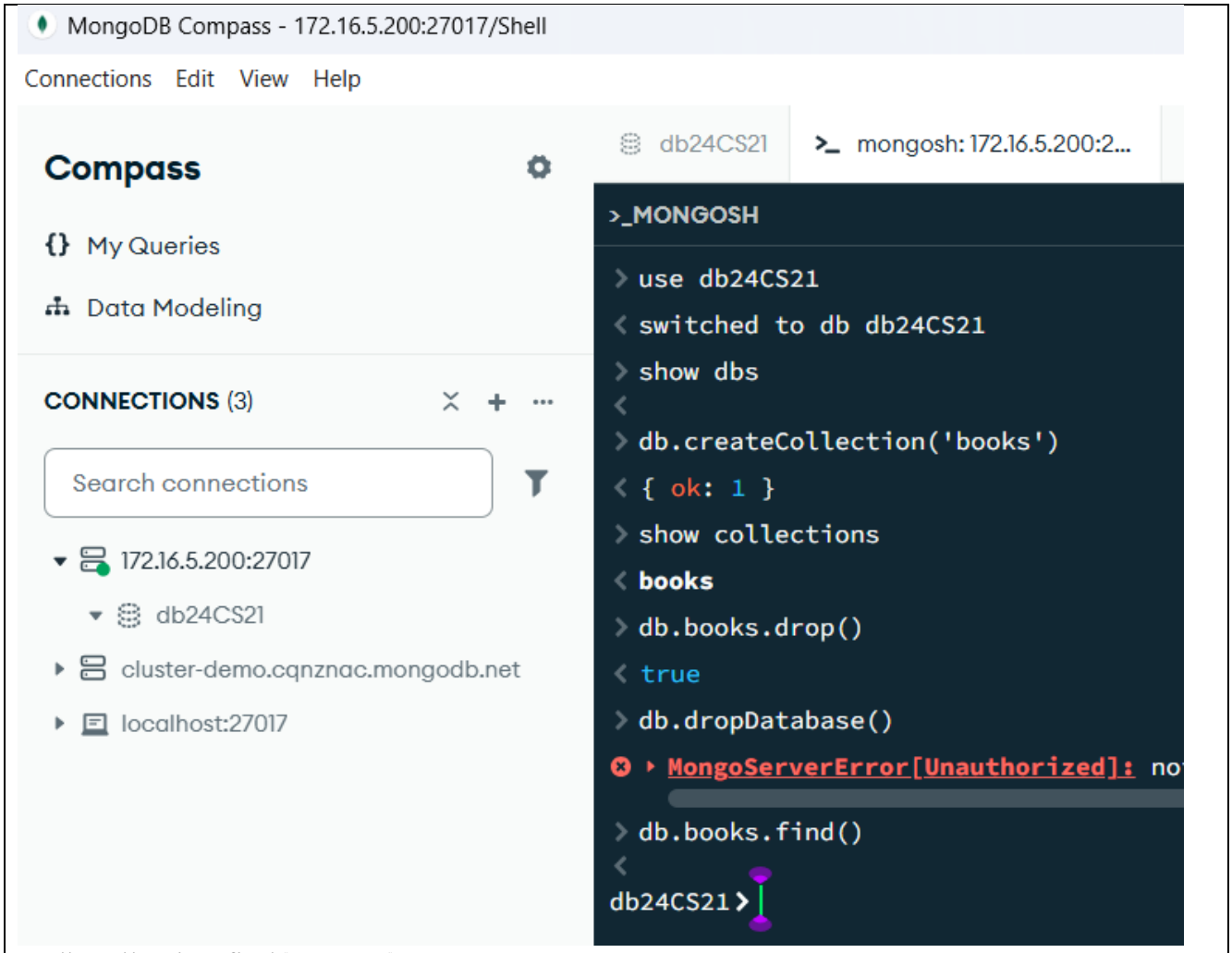
> db.dropDatabase()



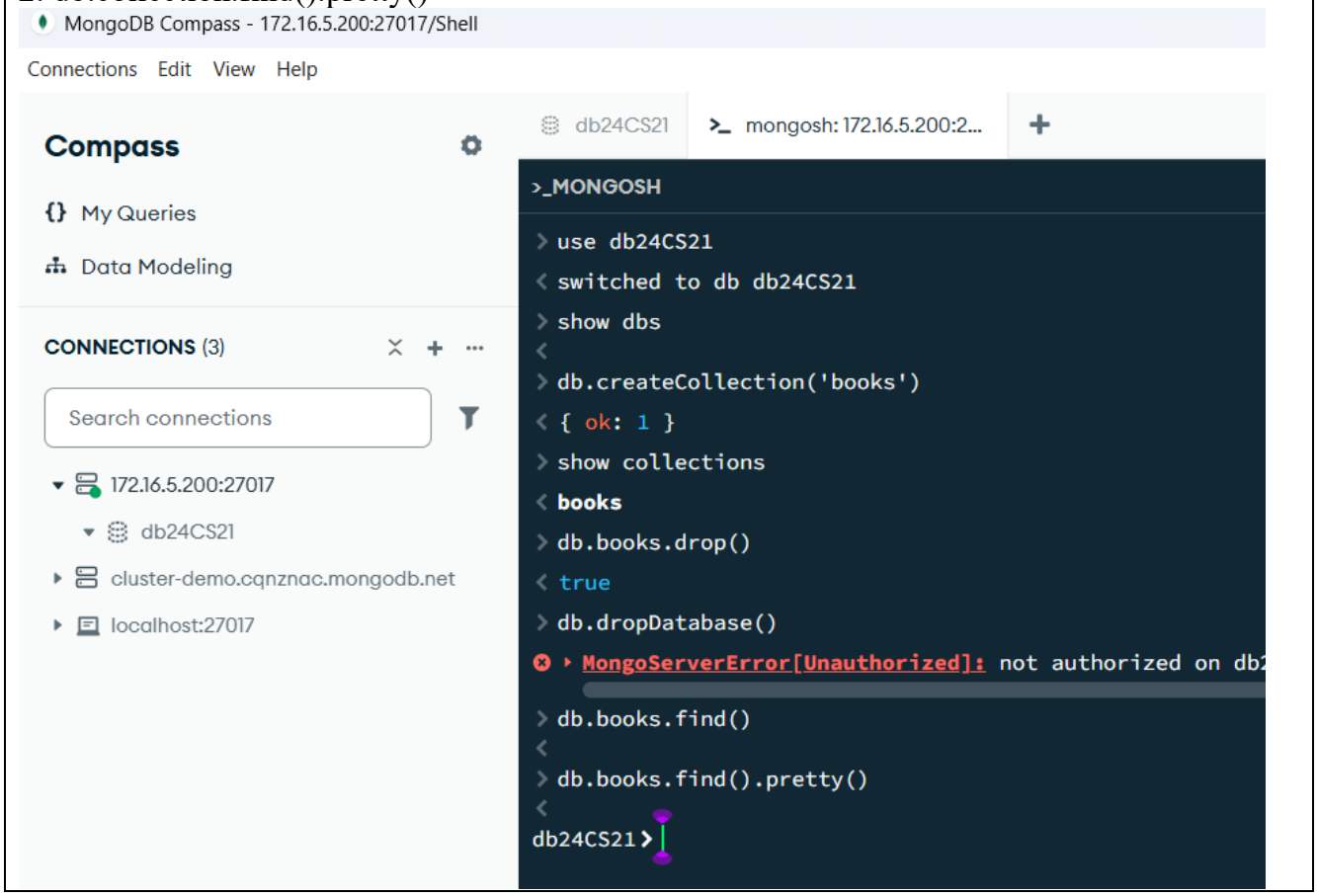
● To retrieve collection

1. db.collection.find()

>db.books.find()



2. db.collection.find().pretty()



● Insert Documents(insert command)

The MongoDB shell provides the following methods to insert documents into a collection:

1. To insert a single document,

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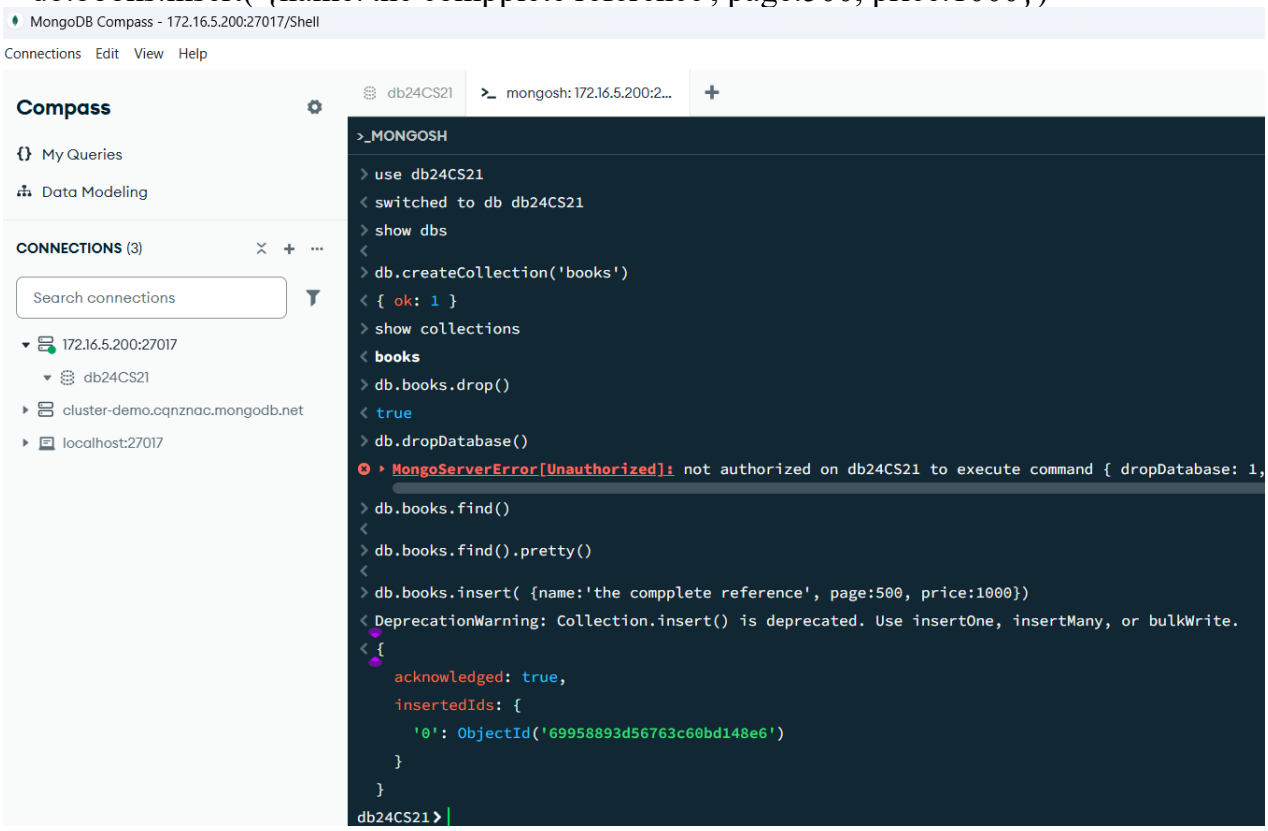
use db.collection.insertOne() or db.collection.insert().

Insert is deprecated since version 3.0 so instead of that we can use insertOne

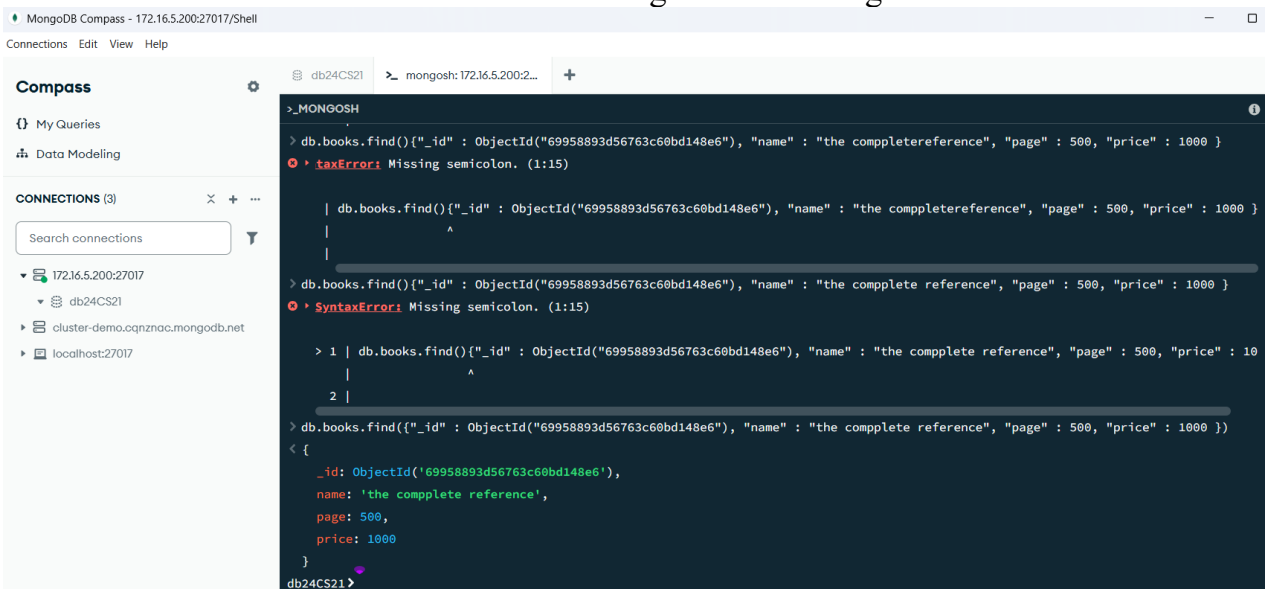
> db.books.insert({name:'the compplete reference', page:500, price:1000})



> db.books.find(){

"_id" : ObjectId("64199568d7fb909707ec850b"), "name" : "the compplete reference", "page" : 500, "price" : 1000 }

//in this command there is an error I have changed it in below given screen shot\\



> db.books.insertOne({name:'the python', page:600, price:800})

MongoDB Compass - 172.16.5.200:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

172.16.5.200:27017

db24CS21

cluster-demo.cqnznac.mongodb.net

localhost:27017

db24CS21 mongosh: 172.16.5.200:2...

>_MONGOSH

```
    name: 'the compplete reference',
    page: 500,
    price: 1000
  }
> db.books.insertOne( {name:'the python', page:600, price:800}){
  "acknowledged" : true,
  "insertedId" : ObjectId("641995c2d7fb909707ec850c")
  }
SyntaxError: Missing semicolon. (1:61)

  > 1 | db.books.insertOne( {name:'the python', page:600, price:800}){
      |                                                               ^
    2 | "acknowledged" : true,
    3 | "insertedId" : ObjectId("641995c2d7fb909707ec850c")
    4 | }
> db.books.insertOne( {name:'the python', page:600, price:800})
< {
    acknowledged: true,
    insertedId: ObjectId('69958accd56763c60bd148e8')
  }
db24CS21>
```

> db.books.find()

MongoDB Compass - 172.16.5.200:27017/Shell

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localhost:27017

db24CS21 mongosh: 172.16.5.200:2...

>_MONGOSH

```
    acknowledged: true,
    insertedId: ObjectId('69958accd56763c60bd148e8')
  }
> db.books.find()
< {
    _id: ObjectId('69958893d56763c60bd148e6'),
    name: 'the compplete reference',
    page: 500,
    price: 1000
  }
  {
    _id: ObjectId('699588ebd56763c60bd148e7'),
    name: 'the compplete reference',
    page: 500,
    price: 1000
  }
  {
    _id: ObjectId('69958accd56763c60bd148e8'),
    name: 'the python',
    page: 600,
    price: 800
  }
db24CS21>
```

2. To insert multiple documents

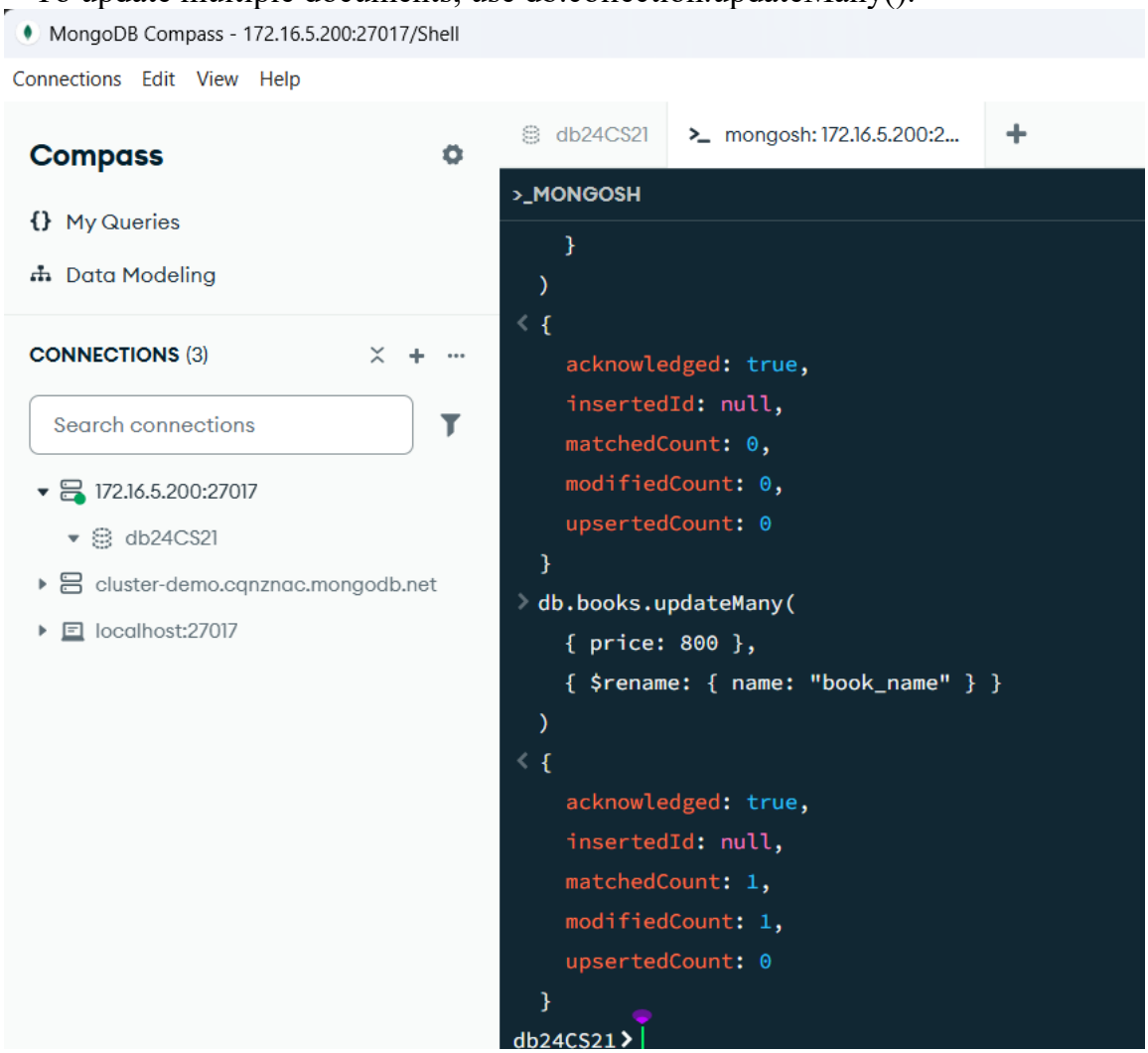
```
db.insertMany.drop()
true
use db.collection.insertMany().
db.books.insertMany([{name:'the compplete reference',
MongoshInvalidInputError: [COMMON-10001] Invalid dat
use db.collection.insertMany().
db.books.insertMany([{name:'the compplete reference',
MongoshInvalidInputError: [COMMON-10001] Invalid dat
db.books.insertMany([{name:'the compplete reference',
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69958c83d56763c60bd148e9'),
    '1': ObjectId('69958c83d56763c60bd148ea'),
    '2': ObjectId('69958c83d56763c60bd148eb')
  }
}
db24CS21>
```

Update Documents(Update command)

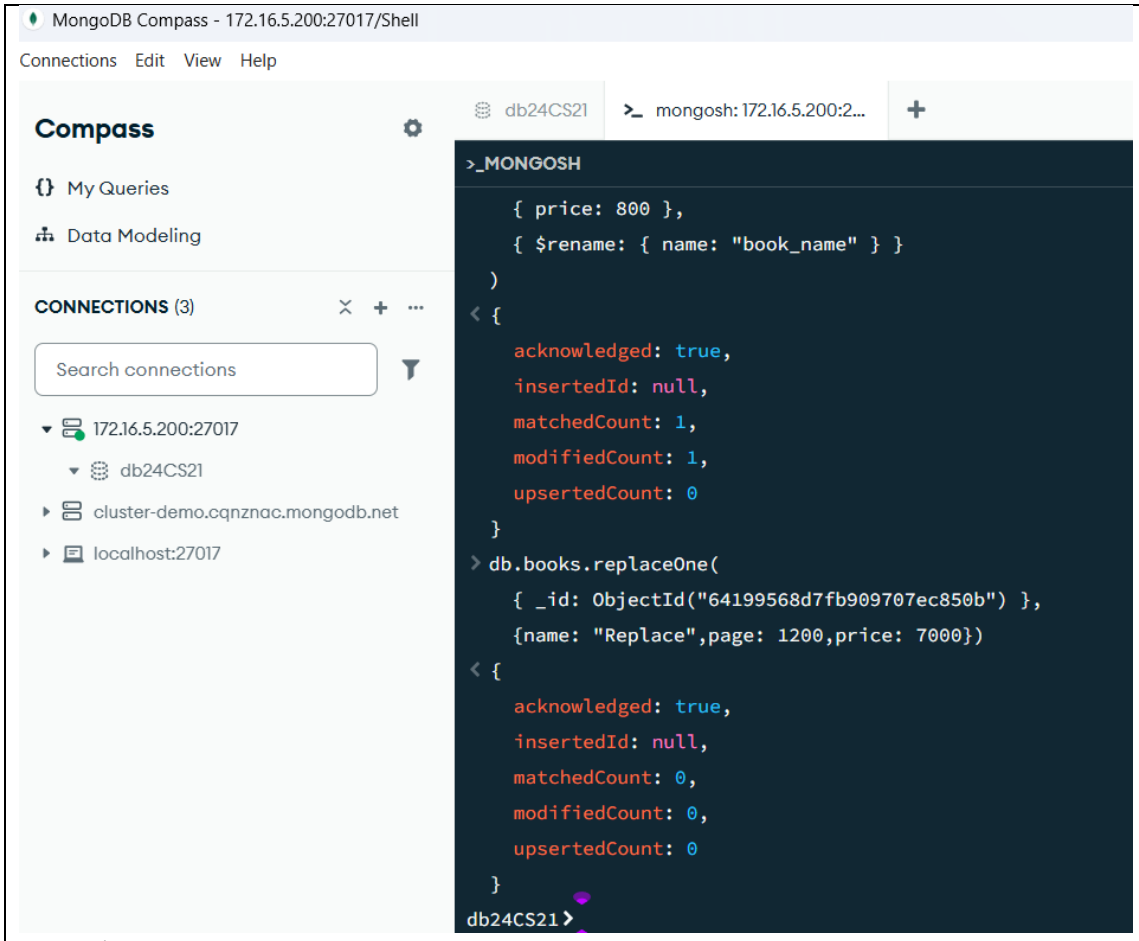
- To update a single document, use db.collection.updateOne(). Or update()

```
MongoDB Compass - 172.16.5.200:27017/Shell
MongoshInvalidArgumentError: Update document requires atomic operators
db.books.updateOne({ _id: ObjectId("69958c83d56763c60bd148e9") },{name: "the co
MongoshInvalidArgumentError: Update document requires atomic operators
db.books.updateOne(
  { _id: ObjectId("64199568d7fb909707ec850b") },
  {
    $set: {
      name: "the complete reference 2",
      page: 8000,
      price: 5000
    }
  }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
db24CS21>
```

- To update multiple documents, use db.collection.updateMany().

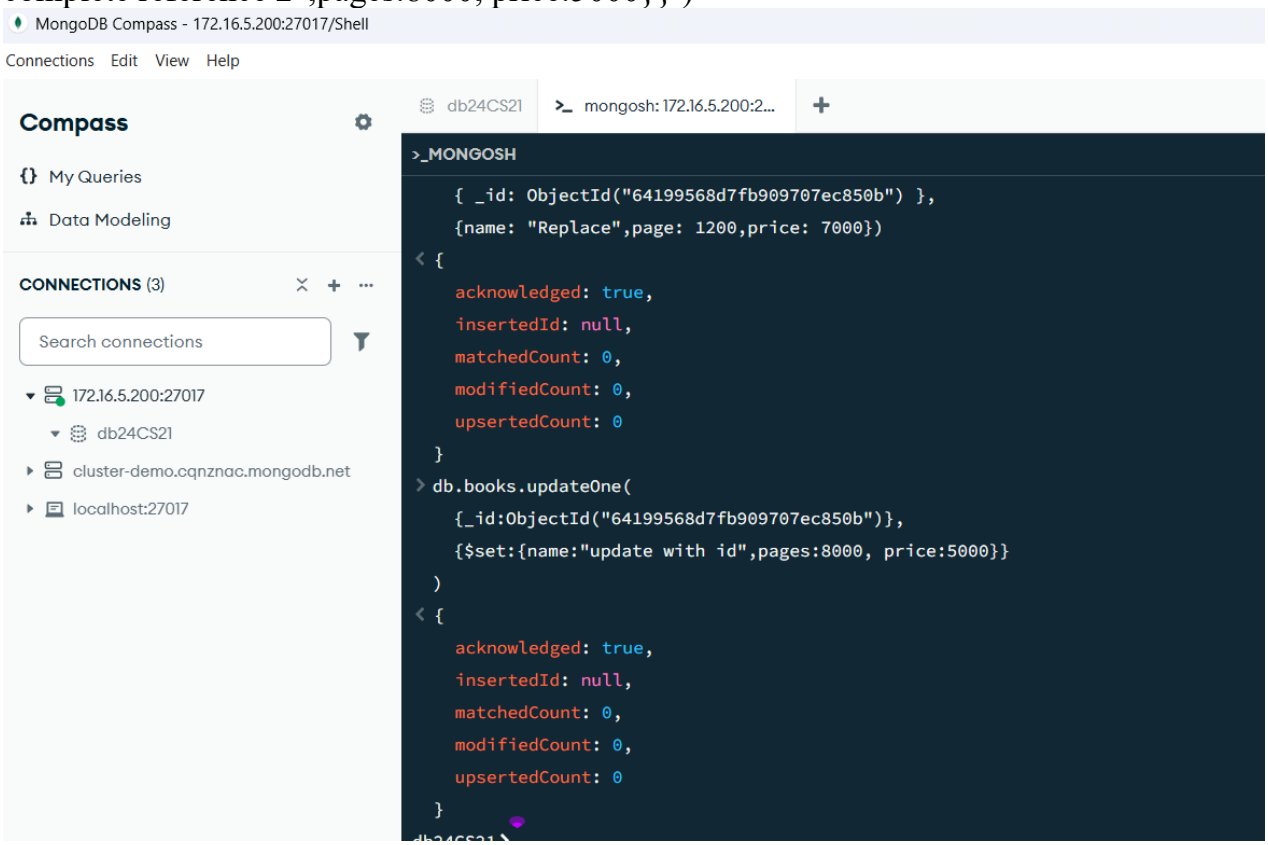


- To replace a document, use db.collection.replaceOne().



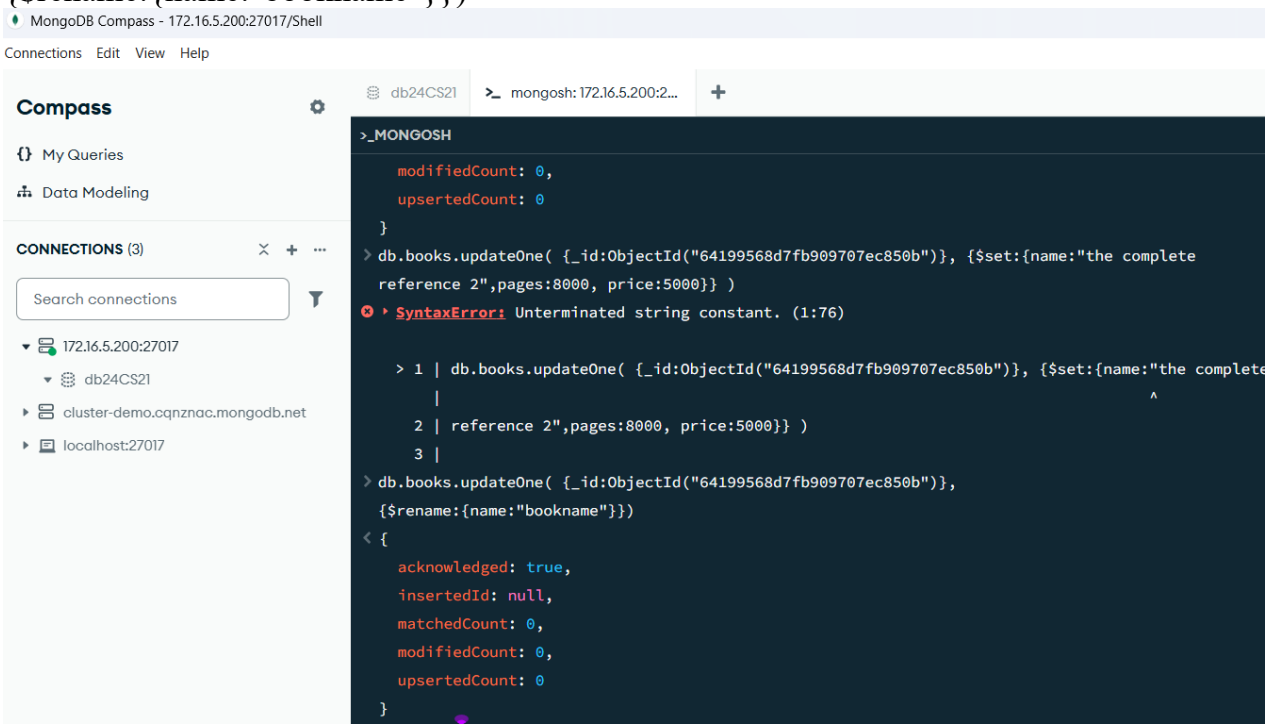
- Update Operators
- Update the document with specific Id

> db.books.updateOne({_id:ObjectId("64199568d7fb909707ec850b")}, {$set:{name:"the complete reference 2",pages:8000, price:5000}})



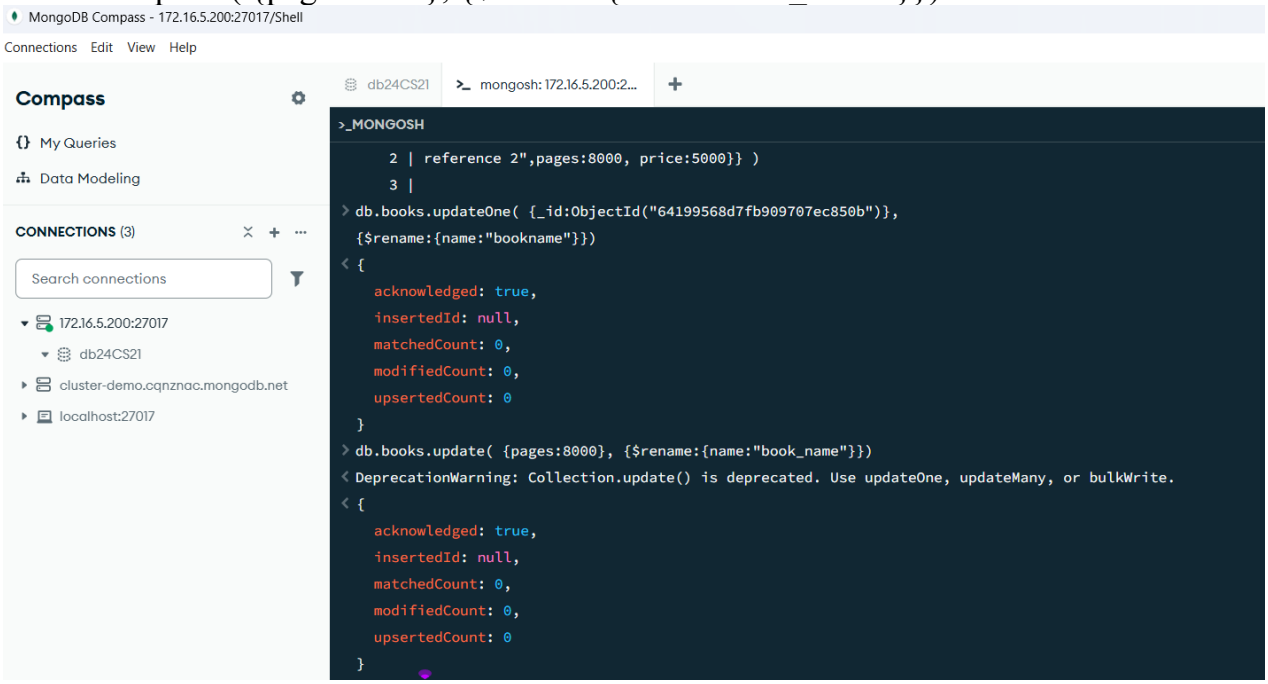
- Update the name to bookname for specific Id

db.books.updateOne({_id:ObjectId("64199568d7fb909707ec850b")}, {$rename:{name:"bookname"}})



- Update the document name to book name where pages=8000

db.books.update({pages:8000}, {$rename:{name:"book_name"}})



- Update all the documents name to book name where price=800

db.books.updateMany({price:800}, {$rename:{name:"book_name"}})

MongoDB Compass - 172.16.5.200:27017/Shell

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localhost:27017

db24CS21 mongosh: 172.16.5.200:2...

>_MONGOSH

```
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.books.update( {pages:8000}, {$rename:{name:"book_name"}})
< DeprecationWarning: Collection.update() is deprecated. Use updat
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.books.updateMany( {price:800}, {$rename:{name:"book_name"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 0,
    upsertedCount: 0
  }
```

- Add 5 in price where price=800

db.books.update({price:800}, {$inc:{price:5}})

MongoDB Compass - 172.16.5.200:27017/Shell

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localhost:27017

db24CS21 mongosh: 172.16.5.200:2...

>_MONGOSH

```
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.books.updateMany( {price:800}, {$rename:{name:"book_name
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.books.update( {price:800}, {$inc:{price:5}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
db24CS21 >
```

Delete Document(delete command)

The MongoDB shell provides the following methods to delete documents from a collection:

● To delete multiple documents, use db.collection.deleteMany().

Delete all those documents which is having title= Titanic

use sample_mflix

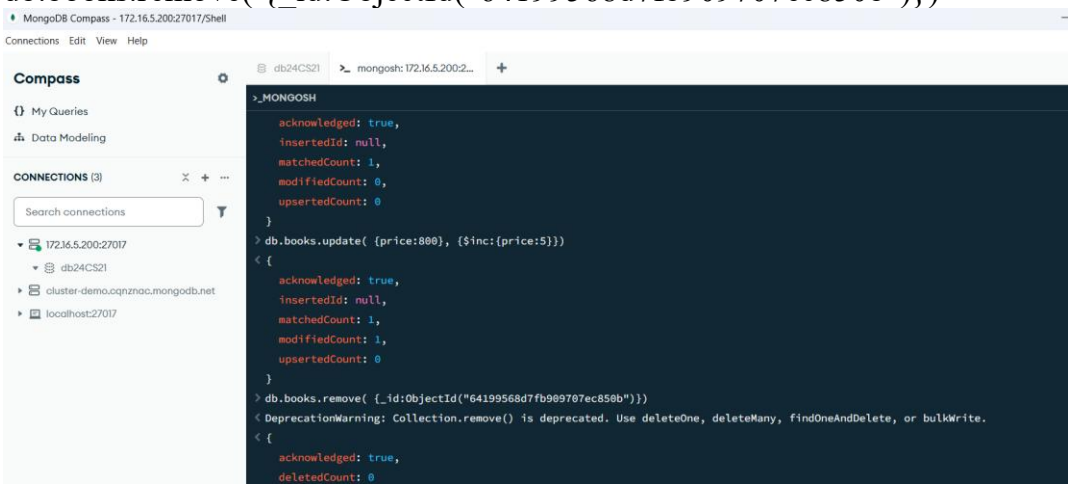
db.movies.deleteMany({ title: "Titanic" })

● To delete a single document, use db.collection.deleteOne(). Or remove()

● remove()

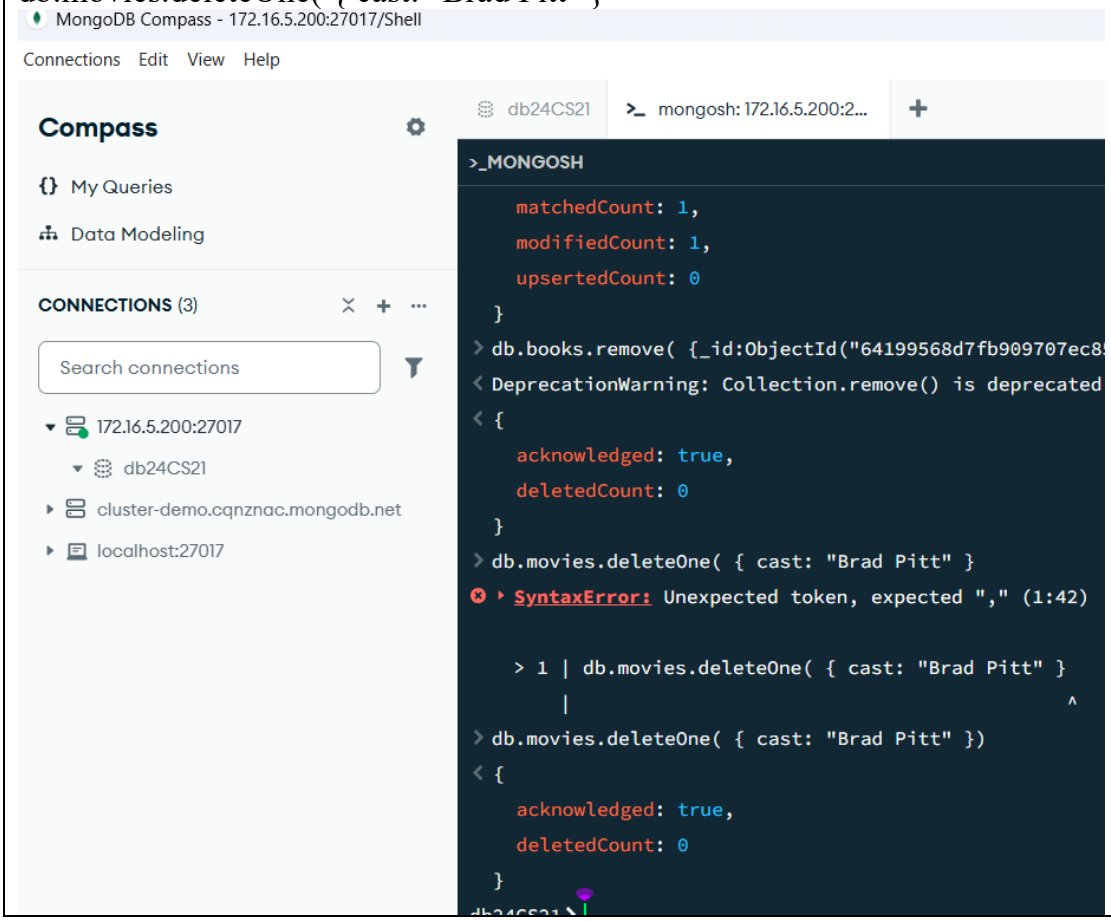
Remove the document with specific Id

db.books.remove({_id:ObjectId("64199568d7fb909707ec850b")})



● deleteOne()

use sample_mflix

Remove the document where cast= Brad Pitt

db.movies.deleteOne({ cast: "Brad Pitt" }